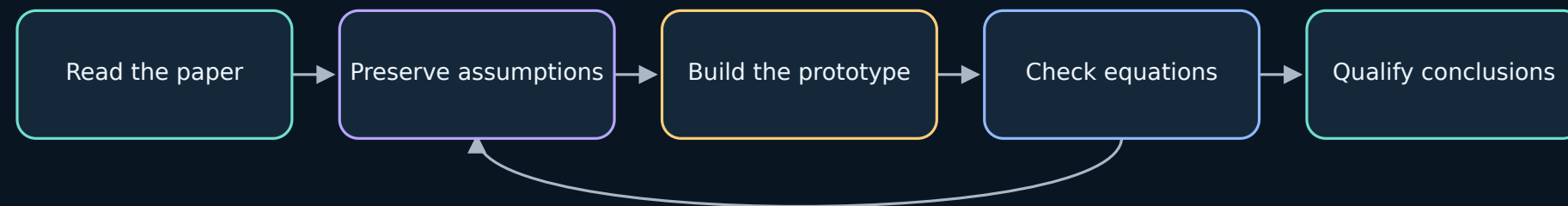


From a paper to a prototype: what needs checking?

AI accelerates implementation; verification determines what we can trust.

- A \$250,000 loss is recorded at a \$100,000 reporting cap.
- “At least” and “exactly” imply different statistical models.
- Build the prototype, test the equations, and qualify the conclusions.

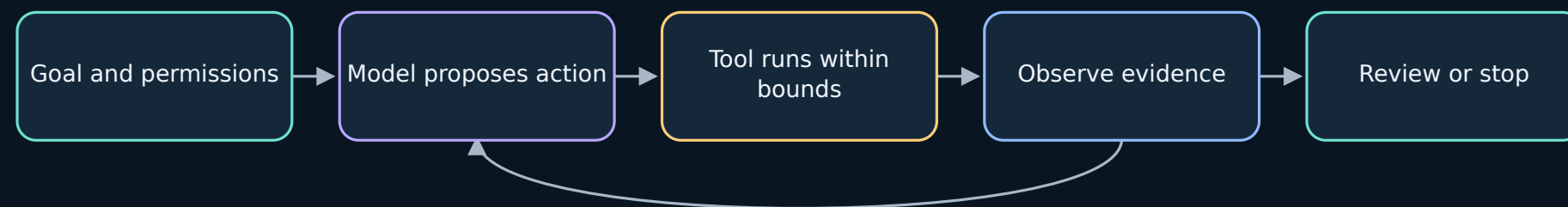


An educational audit: derivatives checked, two claims need correction, and synthetic comparisons do not establish a production winner.

A model becomes useful through a system

Tools, data and verification turn model capabilities into value.

- LLM: generates language and other learned representations.
- Reasoning workflow: spends effort exploring and checking steps.
- Agent: uses tools, observes results and chooses further actions.
- A fluent explanation does not establish correctness.

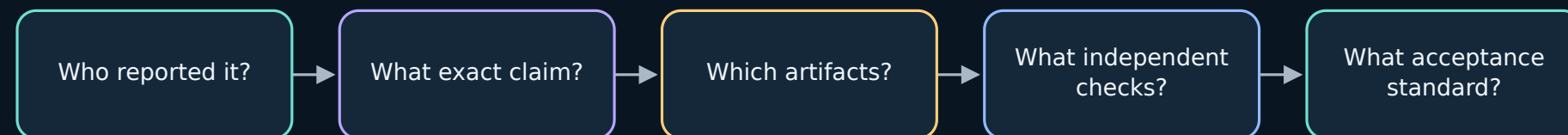


Tool access and a stopping condition are explicit parts of the system.

Read the evidence before repeating the headline

Evaluate the evidence before repeating the claim.

- Ask who made the claim and when.
- Read the exact problem and assumptions.
- Locate the paper, code, data or certificate.
- Separate artifact checking, scientific acceptance and practical usefulness.

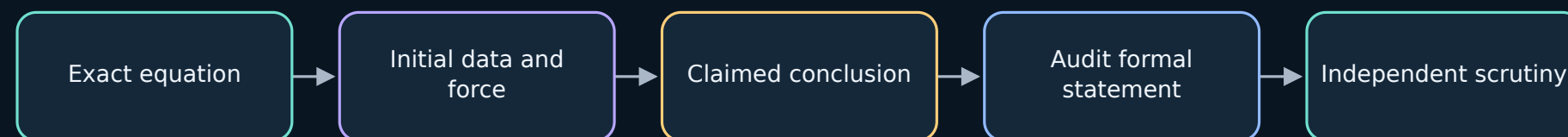


Different disciplines use different acceptance standards. These are questions, not a universal linear ranking.

Navier–Stokes: a precise, attributed frontier claim

Mathematical progress needs precise claims and independent scrutiny.

- OpenAI announced a proposed solution on September 8, 2026.
- The manuscript constructs finite-time blowup with smooth forcing and positive viscosity.
- The exact forced formulation matters.
- Independent acceptance and prize recognition are separate.

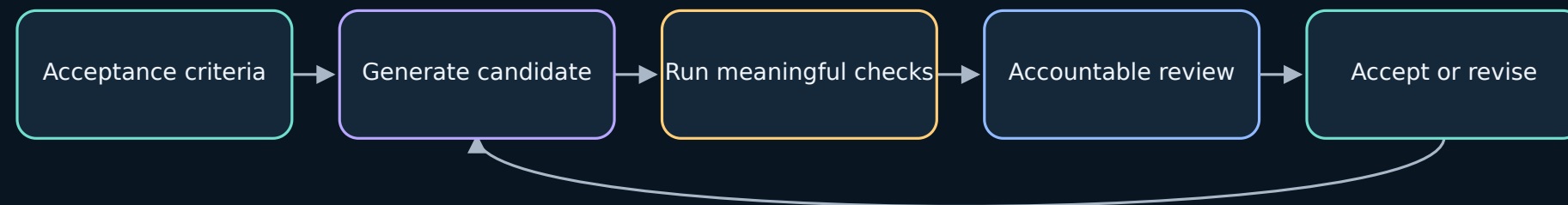


Interpretation guide; this package does not certify a mathematical proof.

Verification is infrastructure for discovery

Build verification into the infrastructure.

- Specify what a valid answer must satisfy.
- Use checks that can reject the proposal.
- Keep provenance, assumptions and failure cases.
- Spend verification effort where errors matter most.

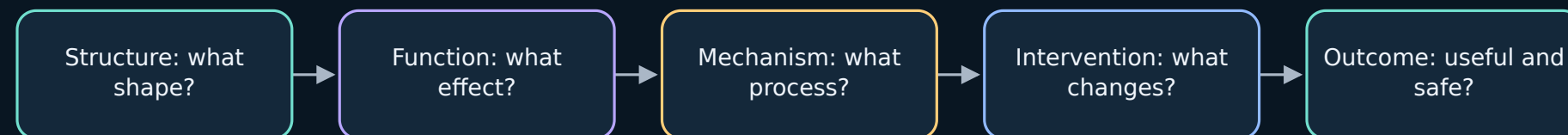


Use checks capable of rejecting a generated result; select the reviewer and criteria before acting.

Scientific AI is a broader ecosystem than LLMs

Scientific AI extends beyond large language models.

- AlphaFold 3 predicts structures of biomolecular complexes.
- Specialized predictors and generators support different tasks.
- LLMs can help connect literature, code and tools.
- Structure, function, mechanism and clinical benefit require different evidence.

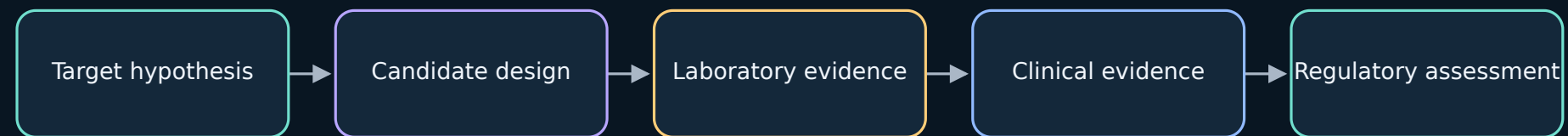


A shape prediction does not by itself identify function, mechanism or therapeutic benefit.

Drug discovery still ends in evidence from people

A candidate molecule still needs clinical evidence.

- AI can support target selection and candidate design.
- Rentosertib has a published phase 2a IPF trial.
- That study involved 71 patients and a 12-week treatment period.
- Its primary objective was safety and tolerability; broader benefit needs further evidence.

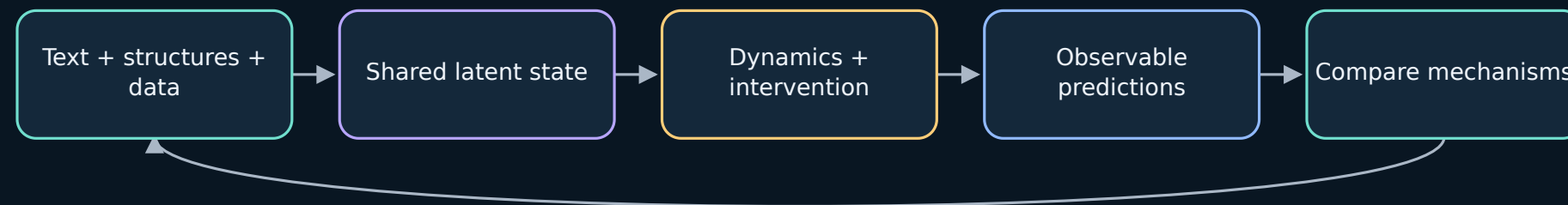


A simplified educational pipeline; stages and regulatory requirements differ by product and jurisdiction.

From prediction and generation to testable mechanisms

Move from prediction and generation to testable mechanistic hypotheses.

- Repository contribution: Shengchao Liu's Wave Intelligence perspective.
- Connect text, structures and measurements through shared representations.
- Model how a system evolves under conditions and interventions.
- Compare alternative mechanisms using observable predictions.

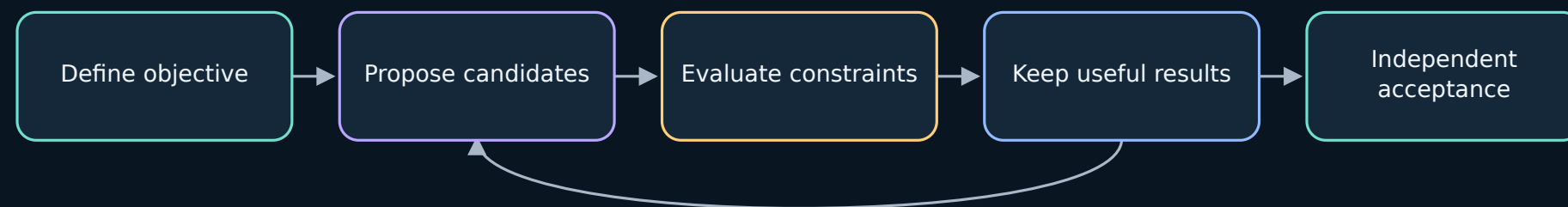


Original explanatory adaptation of Liu's repository perspective, pp. 3-6; a research agenda, not an established validated system.

Research agents need feedback from the world

Research agents need feedback from the real world.

- Coscientist demonstrated tool-connected laboratory workflows.
- AlphaEvolve connects program proposals to automated evaluation.
- The evaluator determines what progress means.
- Simulation success needs an appropriate bridge to reality.

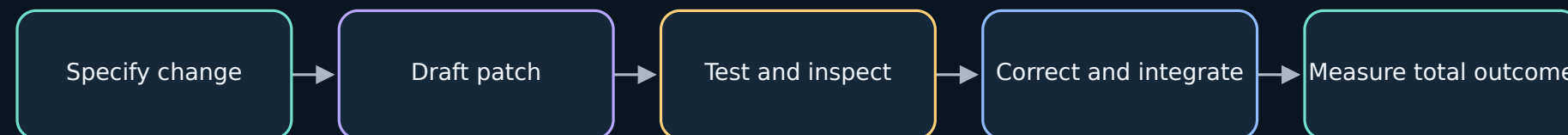


Conceptual abstraction of tool-connected discovery; simulation and physical validation can have different requirements.

In software, measure the whole task

Measure the complete task, including review and rework.

- AI can assist drafting, debugging, tests and code exploration.
- METR's early-2025 experiment found 19% longer task time in its setting.
- The February 2026 follow-up identifies major measurement biases.
- Measure current tools on your own tasks, including review and rework.

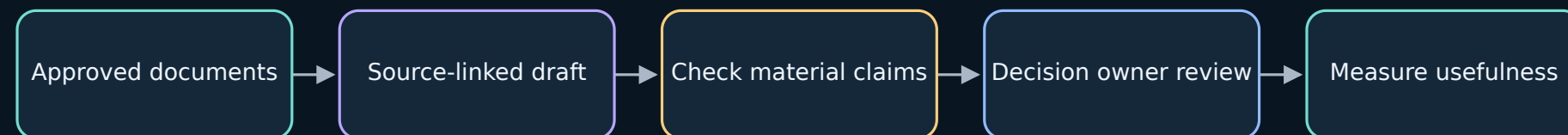


Generation time is only one component. Acceptance includes review and integration.

Knowledge work depends on context and quality

Gains in knowledge work depend on context and quality.

- A support-agent study found heterogeneous productivity gains.
- Its November 2024 manuscript version reports 15% on average.
- That metric is issues resolved per hour in one work setting.
- Transfer requires access to relevant knowledge and quality checks.

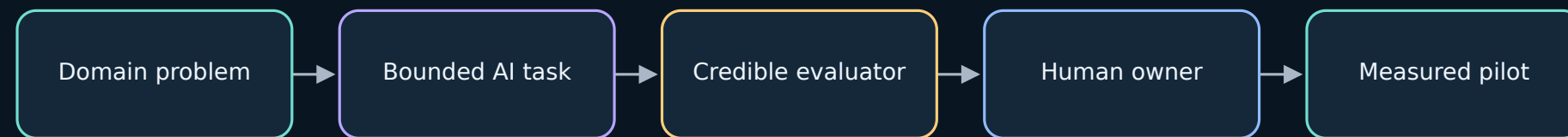


The final decision owner checks material claims and the completeness of the evidence.

Find the opportunity inside your discipline

Combine domain expertise with AI capabilities.

- Engineering: explore alternatives; check constraints and simulation validity.
- Science: organize evidence; design discriminating experiments.
- Software: propose changes; measure integration quality.
- Leadership: redesign workflows; retain accountable owners.

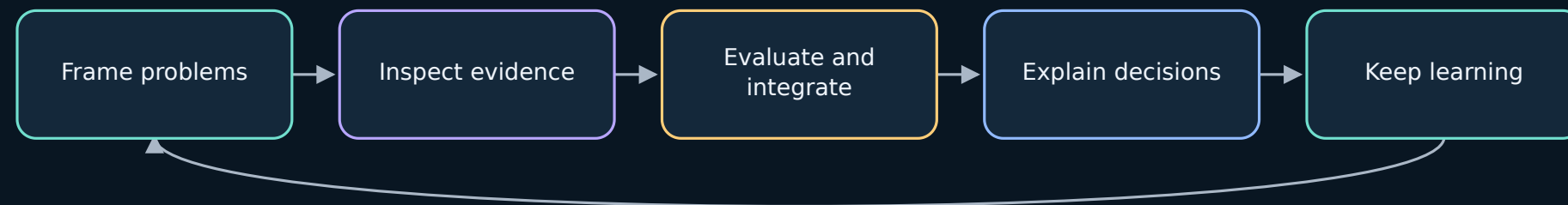


Illustrative suggestions for alumni; no claims about an existing DUT alumni program.

Build the skills that make assistance dependable

Learn to ask, verify, integrate and take responsibility.

- Problem formulation and domain judgment.
- Data literacy, provenance and uncertainty.
- Evaluation, experimentation and tool integration.
- Communication, accountability and continued learning.

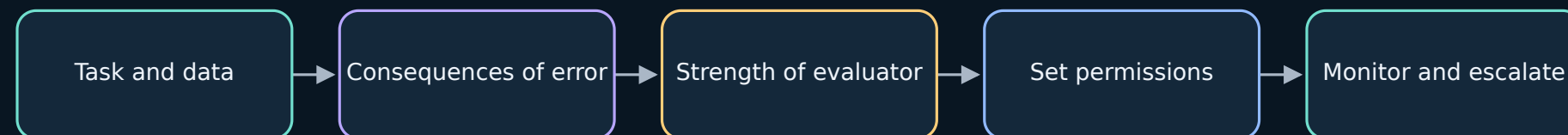


Recommended capabilities to practice together, rather than a forecast of job counts.

Match autonomy to consequences

Match autonomy to consequences and verification capacity.

- Confident errors and unsupported citations.
- Sensitive data and malicious instructions in retrieved content.
- Unequal outcomes, deskilling and unclear responsibility.
- Use bounded permissions, evidence checks and accountable review.

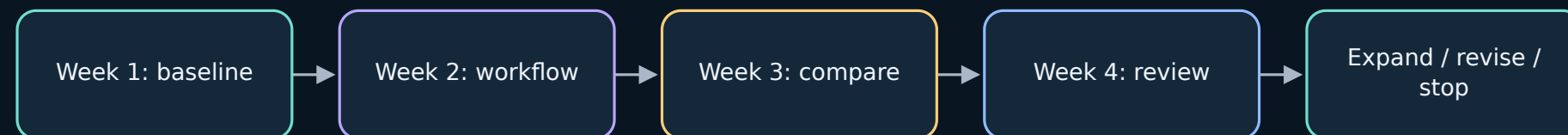


Suggested decision aid, not a numeric risk model or legal determination.

Run one measured pilot in the next 30 days

Run a measurable, reviewable pilot within 30 days.

- Week 1: choose a task, owner and baseline.
- Week 2: define inputs, checks and bounded access.
- Week 3: compare matched tasks and record failures.
- Week 4: decide to expand, revise or stop.

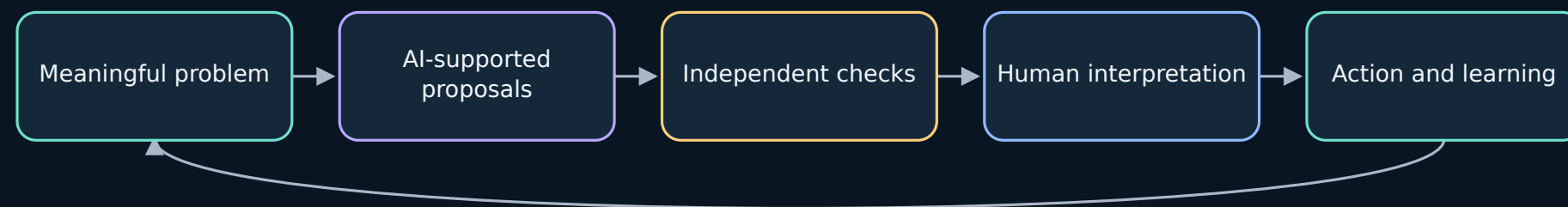


Recommended schedule. Expand, revise and stop are all legitimate outcomes.

More ideas. Stronger evidence. Human responsibility.

Let AI expand exploration, evidence guide action and people retain responsibility.

- Use AI to widen the space of possibilities.
- Use evidence to decide what deserves action.
- Build your career around judgment and reliable execution.
- Panel question: which alumni problem has both value and a credible evaluator?



Conceptual workflow: proposals become useful through evidence and accountable decisions.